

Image Classification

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Image Classification

- ◆ **What is the image classification task, and what are its application scenarios?**
- ◆ What are the challenges in image classification tasks?
- ◆ Is a rule-based method feasible?
- ◆ What is the data-driven image classification paradigm?
- ◆ What are the commonly used evaluation metrics for classification tasks?

What is an image classification task?

- The image classification task is a core task in computer vision, aiming to distinguish images of different categories based on the different features reflected in the image information.

What is an Image Classification Task?

Image Classification: The task of assigning a category label from a fixed set of labels to an input image.

Labels: {Dog, Cat, Truck, Plane, ...}



Dog

What are its applications?



Jade Vine is a rare flower found only in the Philippine rainforest. Its green and blue hues earn it the name “Jade Vine”. Due to habitat loss, it is now an endangered species that can only be cultivated in specific soils.

What are its applications?



The Bedlington Terrier originated in Great Britain. Its primary purpose was to hunt foxes, hares, and badgers. From the late 18th to the early 19th century, it was developed by crossbreeding breeds like the Whippet and Dandie Dinmont Terrier. Through selective breeding, it has evolved into the current height, beauty, speed, and agility, while retaining its original vitality and endurance.

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Semantic Gap

- Crossing the “Semantic Gap”: Establishing Mappings from Pixels to Semantics



What We See

0	3	2	5	4	7	6	9	8
3	0	1	2	3	4	5	6	7
2	1	0	3	2	5	4	7	6
5	2	3	0	1	2	3	4	5
4	3	2	1	0	3	2	5	4
7	4	5	2	3	0	1	2	3
6	5	4	3	2	1	0	3	2
9	6	7	4	5	2	3	0	1
8	7	6	5	4	3	2	1	0

What the Machine Sees

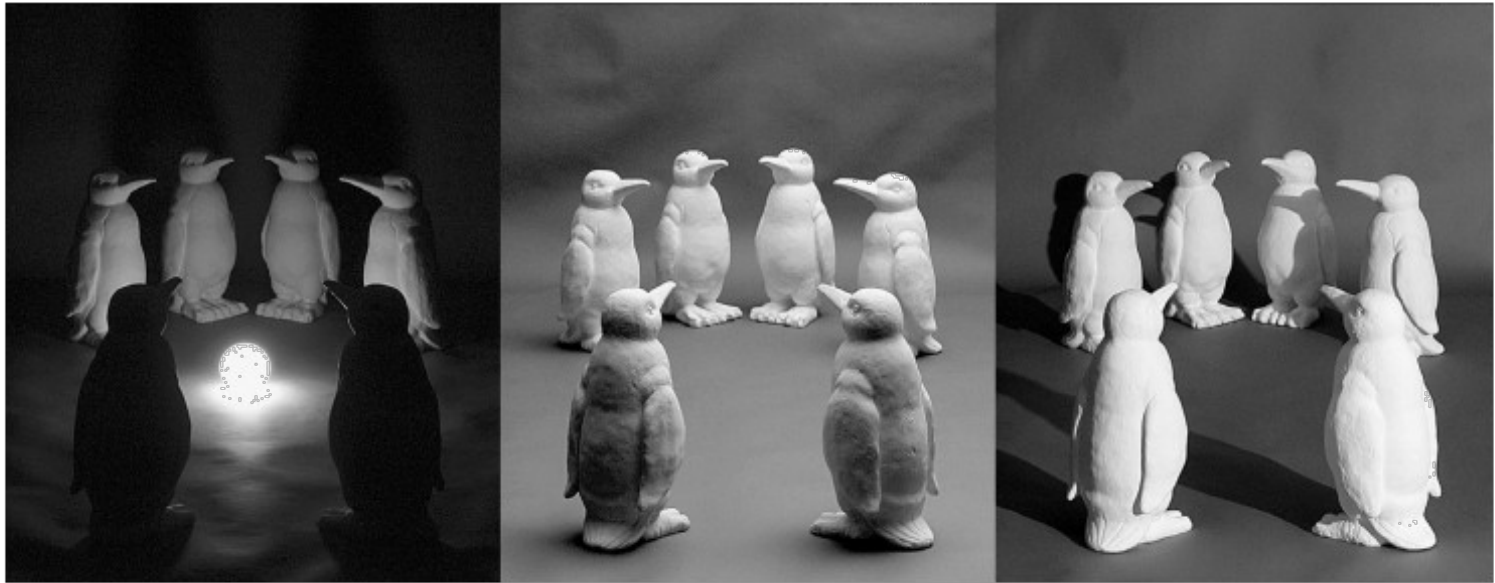
Source: S. Narasimhan

Perspective



Michelangelo 1475-1564

Lighting



Source: J. Koenderink

Scale



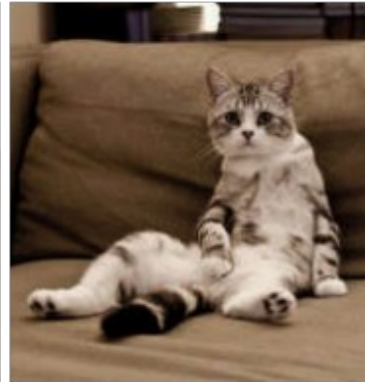
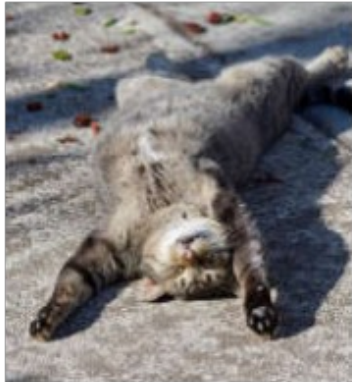
Source: S. Lazebnik

Occlusion



Source: Fei-Fei Li

Deformation

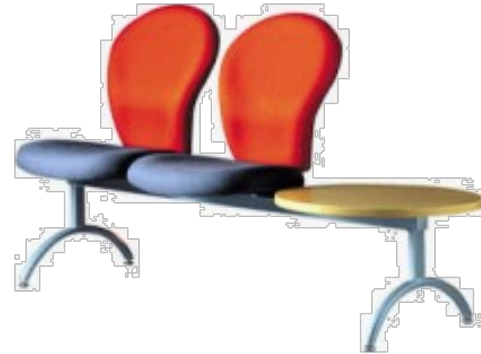


Source: Fei-Fei Li

Background Clutter



Intraclass Deformation



Motion Blur



Source: S. Lazebnik

A wide variety of categories



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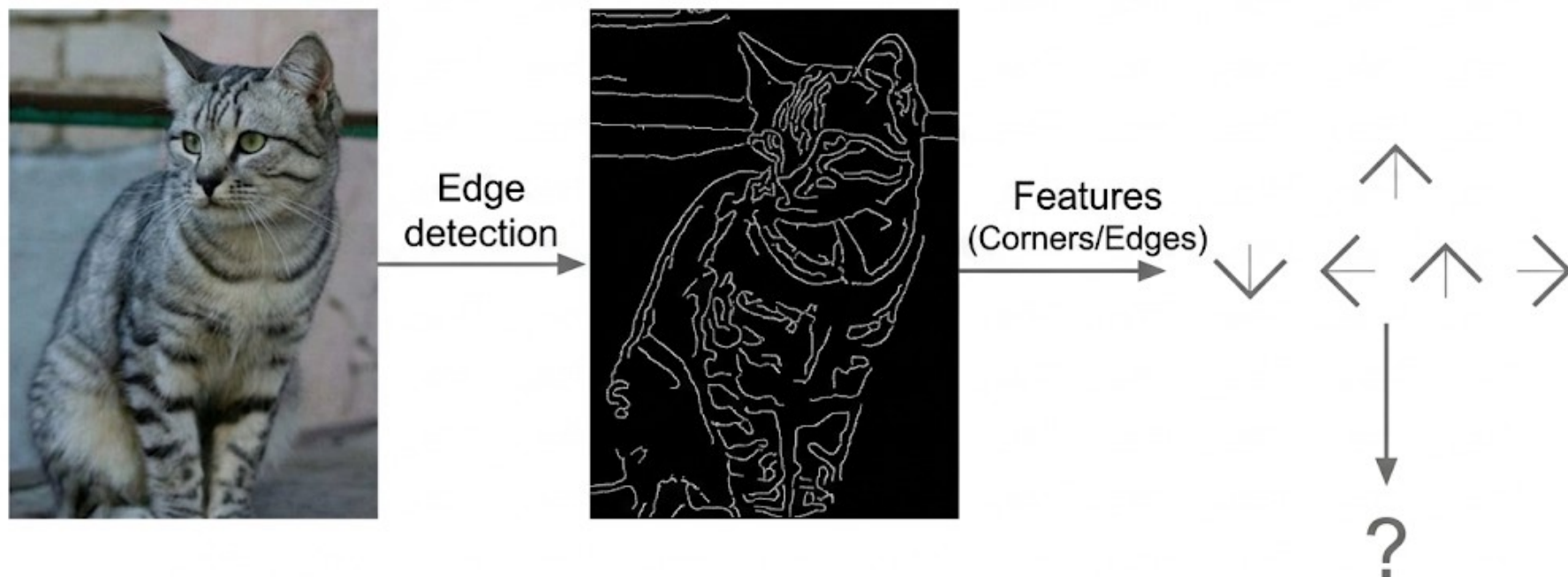
Is a Rule-Based Classification Method Feasible?

```
def classify_image ( image ):  
    # Do something magical here  
    return class_label
```

Identifying cats or other classes using
hard-coding methods

Is a Rule-Based Classification Method Feasible?

Previous attempts:



Is Rule-Based Classification Feasible?

Using **hard-coding** to identify cats or other classes is a **very difficult** task.

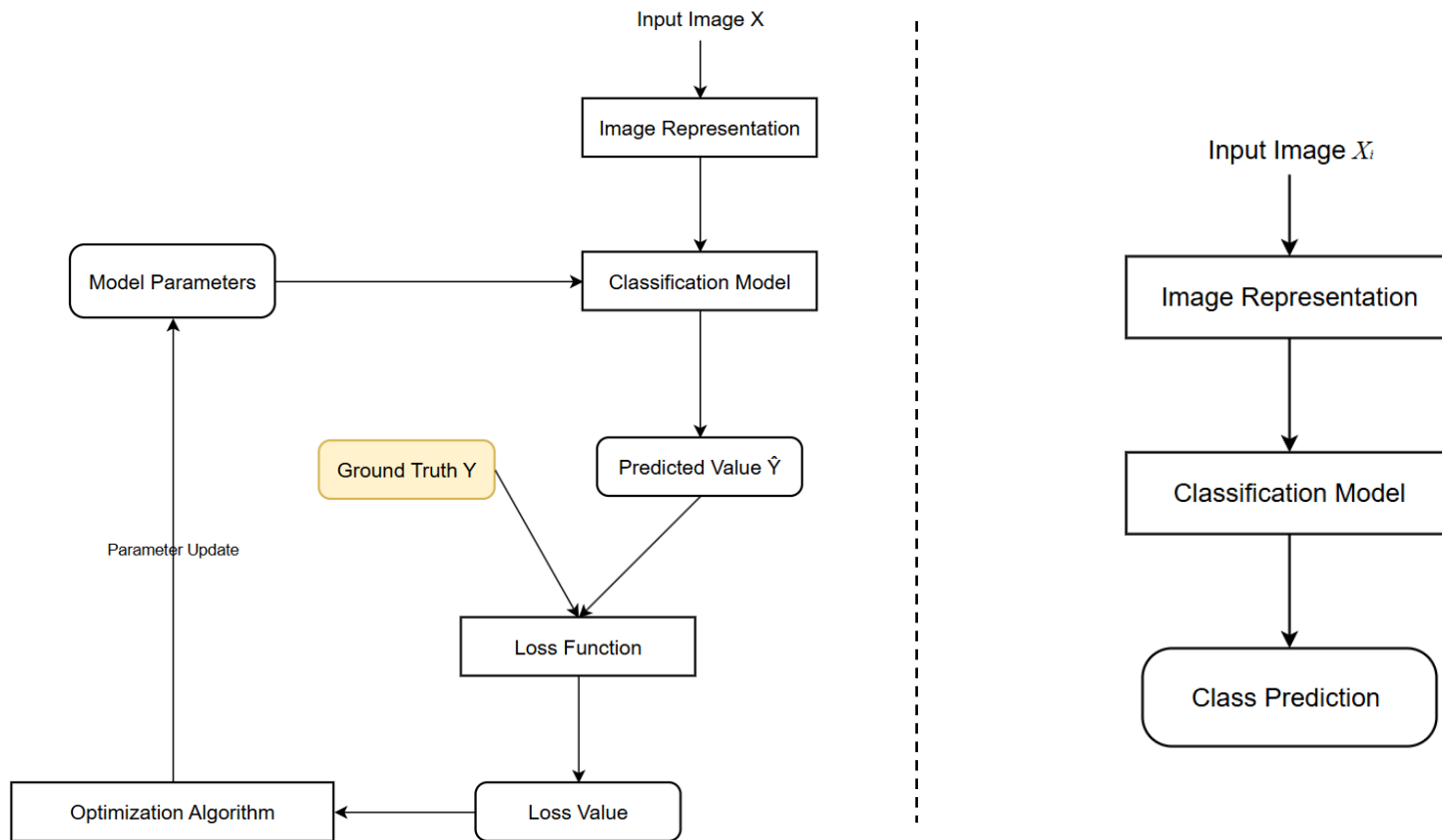
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Data-Driven Image Classification Method

1. Dataset Construction
2. Classifier Design and Learning
3. Classifier Decision-Making

Classifier Design and Learning



Classifier Design and Learning

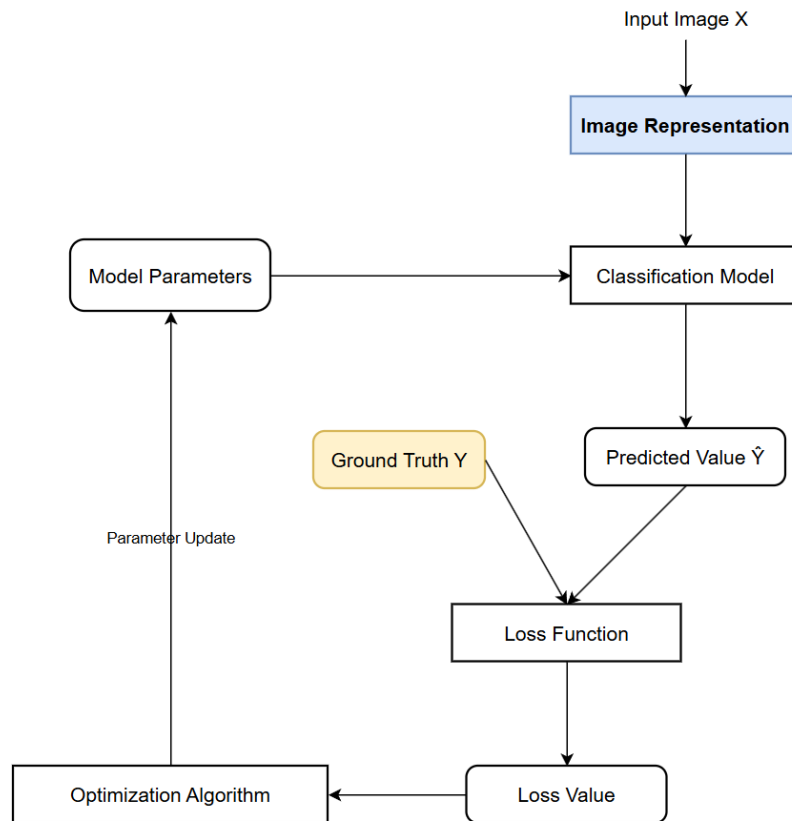


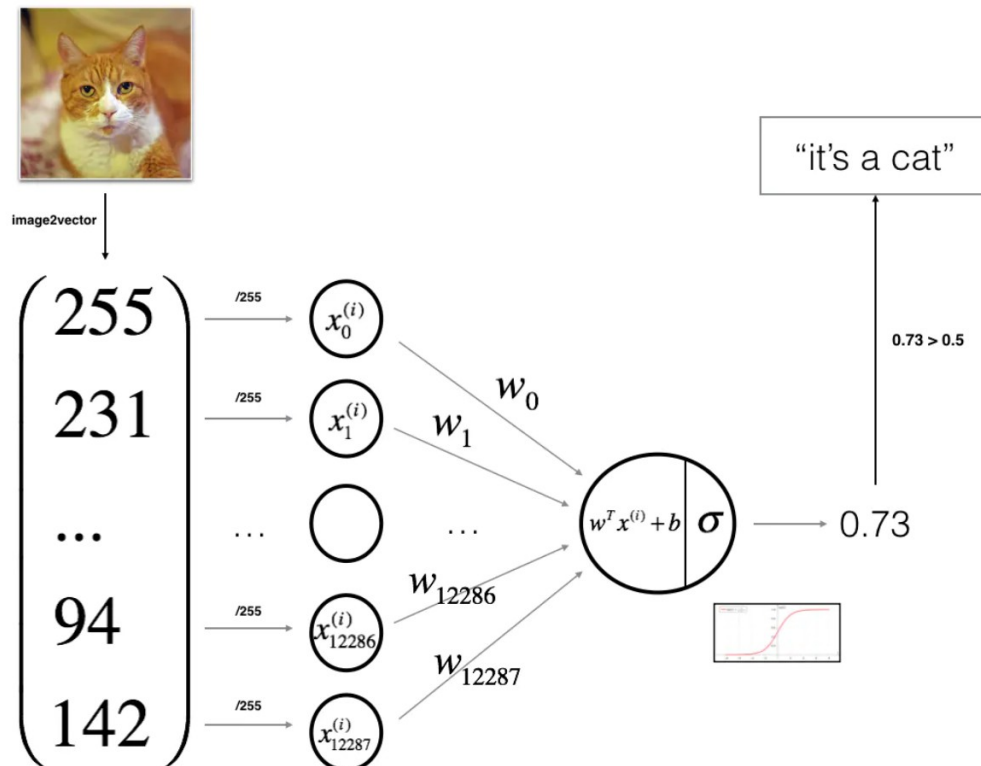
Image Representation

- Pixel Representation
- Global Feature Representation (e.g., GIST)
- Local Feature Representation (e.g., SIFT)

Features + Bag-of-Words Model

Classifier Design and Learning | Feature Representation

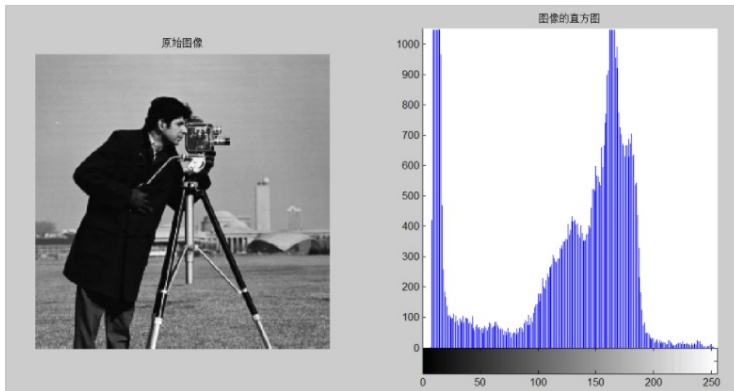
- Pixel Features



Classifier Design and Learning

| Feature Representation

Global Feature Extraction – Image Histogram



$$H = \{h[1], h[2], \dots, h[L]\}$$

$$h[k] = n_k / N_i, \quad k = 0, 1, \dots, L-1$$

- Calculating a color histogram requires dividing the color space into several small color intervals, each of which becomes a bin of the histogram. This process is called color quantization.
- Then, the color histogram can be obtained by counting the number of pixels whose colors fall into each small interval.
- Suppose a color image contains N pixels, and the color space of the image is quantized into L distinct colors, then the color histogram can be calculated using the following formula.

Classifier Design and Learning | Feature Representation

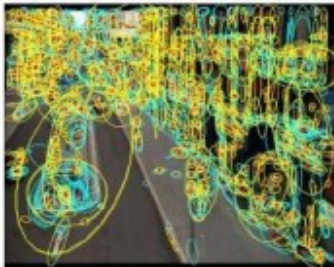
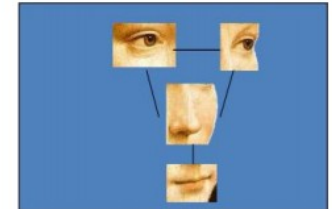
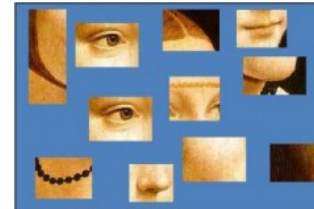
Local Feature Extraction – Sampling Strategy



Interest operators



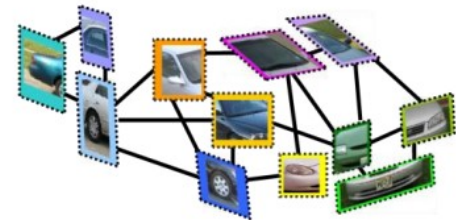
Dense, uniformly



Multiple interest operators



Randomly



Classifier Design and Learning

| Feature Representation

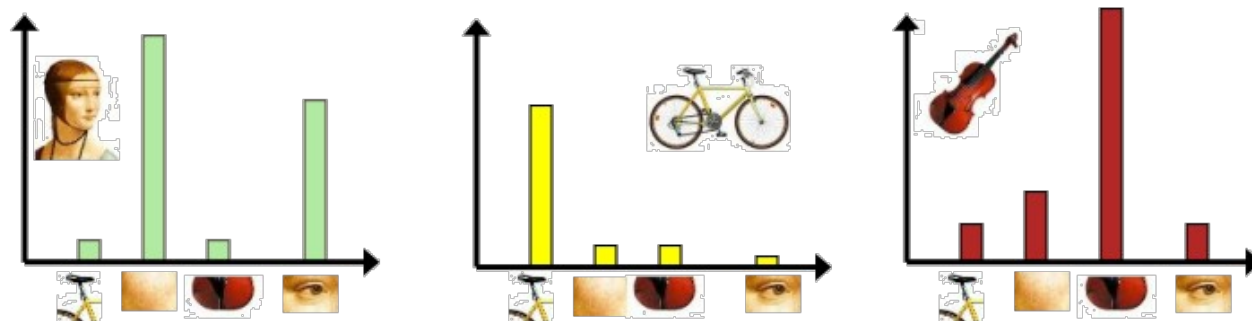
Local Feature Extraction - Bag of Words Model

- Extract Features
- Learn "Visual Words"
- Represent Images by Frequency of "Visual Words"

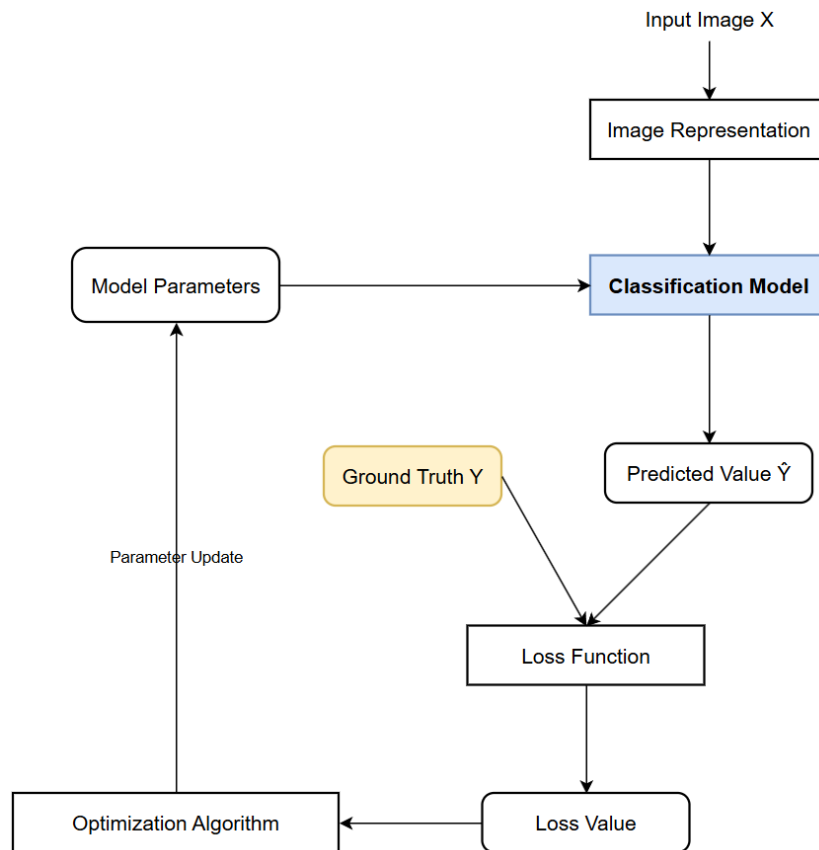
1941-12-08: Request for a Declaration of War

Franklin D. Roosevelt (1933-45)

abandoning acknowledge aggression aggressors airplanes armaments armed army assault assembly authorizations bombing
britain british cheerfully claiming constitution curtail december defeats defending delays democratic dictators disclose
economic empire endanger **facts** false forgotten fortunes france **freedom** fulfilled fullness fundamental gangsters
german germany god guam harbor hawaii hemisphere hint hitler hostilities immune improving indies innumerable
invasion islands isolate **japanese** labor metals midst midway navy nazis obligation offensive
officially **pacific** partisanship patriotism pearl peril perpetrated perpetual philippine preservation privilege reject
repaired **resisting** retain revealing rumors seas soldiers speaks speedy stamina **strength** sunday sunk supremacy tanks taxes
treachery true tyranny undertaken victory **war** wartime washington



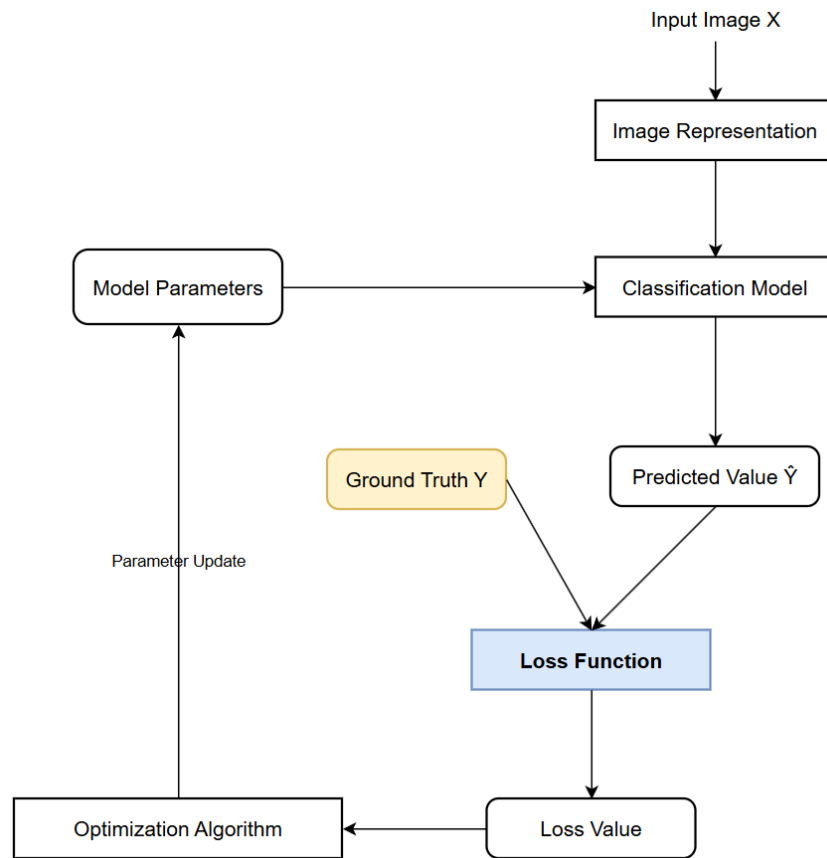
Classifier Design and Learning



Classifiers

- Nearest Neighbor Classifier
- Bayesian Classifier
- Linear Classifier
- Support Vector Machine Classifier
- Neural Network Classifier
- Random Forest Classifier

Classifier Design and Learning

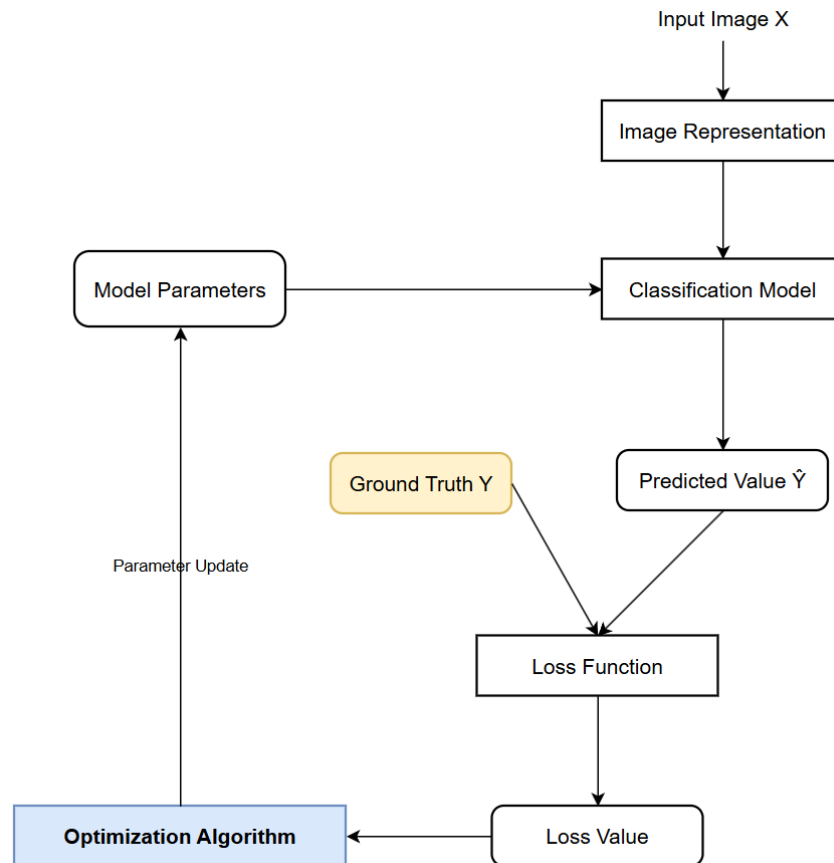


Loss Function

- 0-1 Loss
- Multi-class Support Vector Machine Loss
- Cross-entropy Loss

....

Classifier Design and Learning



Optimization Methods

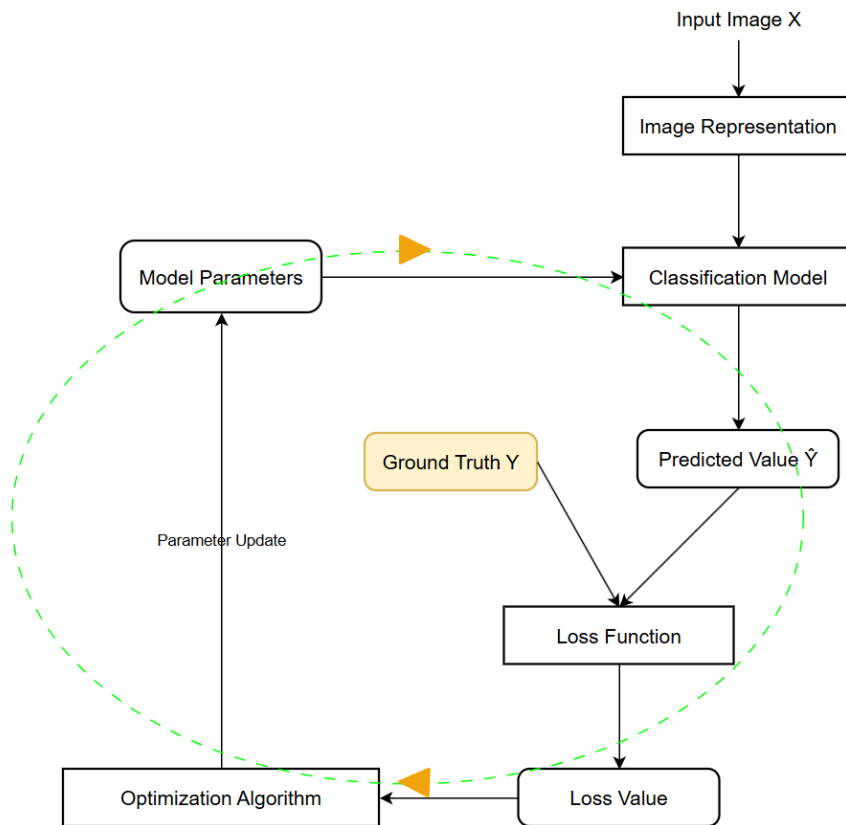
•First-order Methods

- Gradient Descent
- Stochastic Gradient Descent
- Mini-batch Stochastic Gradient Descent

•Second-order Methods

- Newton's Method
- BFGS
- L-BFGS

Classifier Design and Learning



Training Process

- Dataset Splitting
- Data Preprocessing
- Data Augmentation
- Underfitting and Overfitting
 - Reduce Algorithm Complexity
 - Use Weight Regularization Terms
 - Use Dropout Regularization
- Hyperparameter Tuning
- Model Ensembling

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Evaluation Metrics for Image Classification Tasks

- Accuracy = $\frac{\text{Number of correct samples}}{\text{Total number of samples}}$
- Error Rate = 1 - Accuracy

Top-1 Metric and Top-5 Metric



Prediction
→

TOP1

Prediction 1: **Cat** Dog Car Tree Chair

✓

Prediction 2: Dog **Cat** Car Tree Chair

✗

TOP5

Prediction 1: **Cat** Dog Car Tree Chair

✓

Prediction 2: Dog **Cat** Car Tree Chair

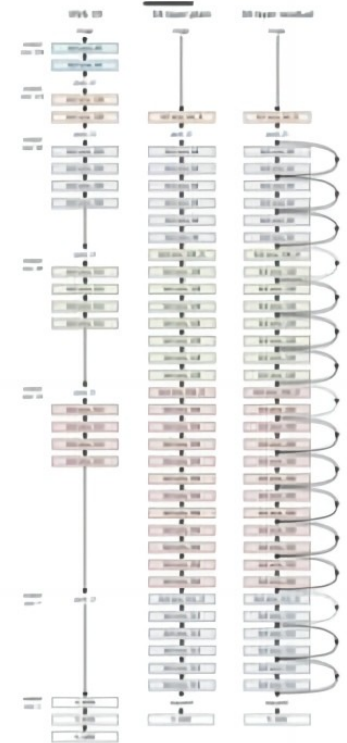
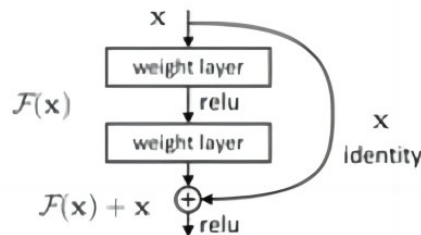
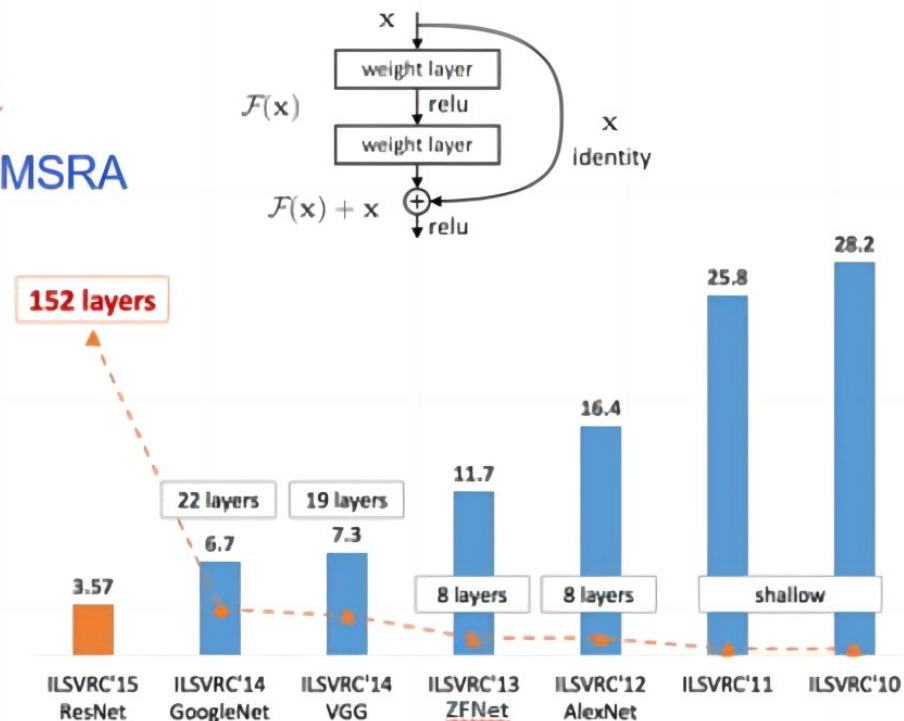
✓

Top-1 Metric and Top-5 Metric

2015年 ResNet

何凯明、孙剑 MSRA

Beat human
benchmark of
5.1% errors



He, K; Zhang, Xu; Ren, S; Sun, J. "Deep Residual Learning for Image Recognition". arXiv:1512.03385, 2015.